Course Project Guidelines

CAI 6107 -- Advanced Machine Learning -- Fall 2026

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Important Notice — FLAIRS-40 Submission Timeline

The course ends in December 2026. The FLAIRS-40 paper abstract deadline is January 22, 2027, and the full paper submission deadline is February 1, 2027. Course grades are finalized before these deadlines.

Teams whose final paper is of sufficient quality are encouraged to submit to FLAIRS-40. Any team wishing to continue refining and submitting their paper after the semester ends is welcome to collaborate further with the instructor. This is entirely voluntary and has no bearing on the course grade, which is based solely on the work submitted by December 1, 2026.

1. Project Overview

The course project is the central deliverable of CAI 6107. Working in teams of 3, students conduct original research on an advanced machine learning topic covered in the course and produce a full-length research paper written to conference standards. The paper follows the format and submission requirements of the Florida AI and Research Symposium (FLAIRS-40), a peer-reviewed international conference indexed in DBLP and Scopus that explicitly welcomes student submissions.

In addition to the paper, teams submit their fully executed Jupyter Notebook with all results visible, the notebook exported as a PDF, and the dataset(s) used. These artifacts allow the instructor to verify and reproduce the experimental work that underpins the paper.

This is genuine research. You identify a gap in existing work, design an experiment to address it, run it on a real dataset, analyze the results, and communicate your findings to a scientific audience.

Project Learning Goals

- Formulate a focused, answerable research question within an ML topic area from the course.
- Review and synthesize relevant literature to identify what is unknown or under-explored.
- Design and execute reproducible experiments on real-world datasets using Python and GitHub Copilot.
- Write a complete full-length research paper that makes and defends an original scientific contribution.
- Give and receive structured peer feedback in the manner of conference reviewing.
- Present research findings in a conference-style oral presentation.

2. Target Venue: FLAIRS-40

FLAIRS-40	Details
Full name	The 40th International Florida AI and Research Symposium
Venue	St. Pete Beach, Florida
Conference dates	May 24-27, 2027
Student papers	Explicitly welcomed and featured at the conference
Paper abstract due	January 22, 2027 (abstract must be registered before full paper can be submitted)
Full paper due	February 1, 2027
Notification	March 9, 2027
Camera ready	April 5, 2027
LaTeX template	Available at flairs-40.info/call-for-papers (Author Template.zip) **Overleaf Access will be provided by Instructor to each tem to work on their project.**

FLAIRS-40	Details
Submission system	EasyChair: easychair.org/conferences/?conf=flairs40
Review process	Double-blind peer review

FLAIRS-40 Topics Relevant to This Course

The following FLAIRS-40 topic areas directly align with CAI 6107 course topics and are confirmed in the official Call for Papers:

- Machine Learning
- Data Mining
- Multimodal AI
- Healthcare Informatics
- Cybersecurity (anomaly detection applications)
- AI for Scientific Discovery (biclustering, graph mining applications)
- Artificial Intelligence in Education
- Edge AI (federated learning applications)

3. Paper Requirement

All teams are required to submit a full paper. There is only one paper type accepted for this course project.

Type	Length	Requirements
Full paper	Up to 6 pages + references	Mature novel research with a clear research question, complete experiments, baseline comparisons, and measurable contribution. Accepted at FLAIRS-40 for a 20-minute oral presentation.

Formatting Requirements

- Use the official FLAIRS-40 LaTeX template (Author Template.zip from flairs-40.info/call-for-papers). For this we wll work in overleaf prvided by faculty.
- Papers must be compiled to PDF. Word submissions are not accepted from Milestone 3 onward.
- Double-blind: submitted PDFs must not include author names, affiliations, or acknowledgments.
- References are excluded from the page count. Appendices are permitted after references but may not be reviewed.
- All figures must have captions and be readable at print size (minimum 9pt font within figures).
- The AI Usage Statement is appended after references and excluded from the page count.
- The Team Contribution Statement is appended after the AI Usage Statement and excluded from the page count.

Required Paper Structure

Section	Approx. length	What it must contain
Abstract	150-200 words	Problem; approach; key result; significance. Written last, appears first. No citations. No undefined abbreviations.
1. Introduction	~0.75 page	Motivation and context; problem statement; explicit list of contributions; roadmap of remaining sections.
2. Related Work	~0.75 page	Thematic groups of prior work (not chronological); gap analysis; at least 10 peer-reviewed citations; explanation of what each group of prior work does not address that this paper does.
3. Methodology	~1.25 pages	Problem formulation with notation; proposed approach described precisely enough to replicate; algorithm or architecture diagram; justification of design choices.

Section	Approx. length	What it must contain
4. Experiments	~0.75 page	Dataset(s) with source, size, and splits; at least one baseline method; evaluation metrics with definitions; train/validation/test protocol; reproducibility details.
5. Results & Discussion	~0.75 page	Results tables and/or figures with captions; comparison against baselines; analysis of why results occur; limitations; ablation study if applicable.
6. Conclusion	~0.25 page	Summary of key findings (not a copy of abstract); concrete future directions; honest statement of scope and limitations.
References	Unlimited	All in-text citations have matching entries; formatted by FLAIRS LaTeX template; at least 10 peer-reviewed sources.

4. Code, Notebook & Dataset Submission

In addition to the research paper, every team submits three supplementary deliverables with their final submission. These allow the instructor to verify and fully reproduce all experimental results reported in the paper.

Jupyter Notebook (fully executed)

- The notebook must be fully executed, all cells run in order from top to bottom with all outputs, figures, and results visible.
- The notebook must be self-contained: someone cloning the repository and running the notebook should be able to reproduce your results with no additional setup beyond installing the listed packages.
- Include a requirements cell at the top that lists all Python package dependencies and versions (e.g. pip install sklearn==1.3.0 pandas==2.0.0).
- Code comments must explain the purpose of each major block. GitHub Copilot-generated code must be reviewed and understood by the team.
- Notebook filename: TeamName_notebook.ipynb

Notebook PDF

- Export the fully executed notebook as a PDF (File > Download as > PDF via nbconvert or Jupyter/Colab export).
- The PDF must show all code cells and all output cells including tables, figures, and printed results.
- This PDF is the document the instructor uses for grading, it must be complete and readable.
- Notebook PDF filename: TeamName_notebook.pdf

Dataset

- Submit the dataset(s) used in your experiments. If the dataset is publicly available and too large to upload, include a download script or a link with exact version/access date.
- If data was collected, scraped, or custom-assembled by the team, include the full raw dataset.
- Include a brief README.txt describing the dataset: source, size, features, target variable, and any preprocessing applied before it reaches the notebook.
- Dataset folder: TeamName_data/ (or a compressed TeamName_data.zip)

What makes a strong notebook?

- Clear section headers matching the paper's Experiments and Methodology sections.
- All baseline comparisons implemented and run in the same notebook.
- Visualizations that correspond exactly to figures in the paper.
- Final results table(s) that match the numbers reported in the paper exactly.
- No silent failures, if a cell raises an error, it must be fixed before submission.

5. Teams

Teams consist of 3 students. Solo work is not permitted. Every team member is expected to contribute substantially to all aspects of the project, writing, coding, experiments, and presentation.

A Team Contribution Statement must be appended to all submissions from Milestone 3 onward. The statement lists each member's specific contributions by section and task. Significant documented imbalances in contribution may result in adjusted individual grades at the instructor's discretion.

6. Topic Selection

Your research question must connect to at least one of the course topic areas. Any of the 13 areas below is a valid choice, there is no restriction to only the topics newest to the field.

Data pre-processing & feature selection	Anomaly detection	Ensembles & committee machines
Multilabel classification	Biclustering	Recommender systems
Semi-supervised learning	Reinforcement learning	Transfer learning / meta-learning
Federated learning	Multimodal learning	Unstructured data mining
Graph & network mining		

Choosing Your Topic: Timing and Novelty Tradeoffs

Topics taught earlier in the semester give your team a longer runway before each milestone, but tend to be more heavily studied in the literature, which raises the bar for finding a genuinely novel angle. Topics taught later carry more natural novelty, since less has been published, but compress the time between the lecture and the milestone that depends on it. Choose deliberately, not by default.

What makes a strong project?

- Apply a known method to a new dataset or domain where it has not been tested.
- Combine techniques from two topic areas and study the interaction.
- Propose a modest modification to an existing algorithm with empirical justification.
- Conduct a reproducibility study that replicates and meaningfully extends a published result.

What makes a weak project?

- Re-implementing an existing algorithm on the same benchmark used in the original paper with no new insight.
- A literature review with no experiments.
- A project with no baseline comparisons.

7. Milestones & Grading

The project is worth 40 course points total, distributed as follows. Late submissions lose 2 points per day. Milestones more than 5 days late receive zero credit. The final submission deadline is firm.

Component	Points	Description
M1 -- Team & topic registration	2	Team list (2-3 names) + topic area on Canvas.
M2 -- Research proposal	3	2-page document: research question, gap, dataset(s), method, evaluation plan.
M3 -- Introduction & Related Work	5	Sections 1-2 in FLAIRS LaTeX; 10+ peer-reviewed citations; thematic organization.
M4 -- Methodology & Experiments draft	5	Sections 3-4 in LaTeX; replicable method; baselines defined; dataset stats; metrics justified.
M5 -- Complete paper draft	5	Full paper all sections in FLAIRS format; submitted for peer review exchange.

Component	Points	Description
Peer review	5	Structured written review of assigned team's draft; graded on specificity and constructiveness.
Final paper (PDF)	10 (see distribution in section 8)	Full 6-page paper in FLAIRS format; graded on novelty, soundness, clarity, and significance.
Jupyter Notebook (fully executed)	3	All cells run top-to-bottom; results visible; self-contained; commented code.
Dataset submission	2	Dataset + README; results in paper match notebook output exactly.
TOTAL	40	

8. Final Paper Grading Rubric (10 points)

The final paper (10 of the 40 project points) is graded against the following criteria. The same criteria apply when teams conduct their peer review of another team's draft.

Section / Criterion	Pts	Full credit	Partial / no credit
Abstract	1	States problem, approach, key result, and contribution clearly. 150-200 words. No citations.	Vague; missing results; exceeds length.
Introduction	1.5	Clear motivation; explicit research question; numbered contribution list; structure outline.	Background only; contributions not listed; no structure outline.
Related work	2	Thematic organization; gap analysis; 10+ peer-reviewed citations; positions own work.	Chronological summaries; no gap; fewer than 8 citations.
Methodology	2	Replicable description; all notation defined; diagram provided; choices justified.	Vague; undefined notation; no diagram; unjustified choices.
Experiments	1.5	Dataset described with stats; at least one baseline; metrics defined; reproducible protocol.	No baselines; metrics undefined; dataset underspecified.
Results & Discussion	1.5	Tables/figures with captions; baseline comparisons; analysis of why results occur; limitations.	Numbers without analysis; no comparison; limitations ignored.
Conclusion & writing	0.5	Summarizes findings; concrete future work; FLAIRS format; grammatically correct.	Restates abstract; vague future work; wrong format.

9. Presentation Rubric (part of course grade, not project points)

The conference-style presentation is graded separately as part of the course grade (10 points in the course grading breakdown). Each team presents once, during the final exam slot, Tue Dec 01, 3:00-5:15 pm.

Criterion	Pts	Strong	Weak
Motivation & problem setup	2	Problem clearly motivated within 2 minutes; research question explicit; audience understands why it matters.	Motivation unclear; question never stated.
Technical clarity	3	Method explained correctly; key equations or diagrams shown; results interpreted, not just displayed.	Method incorrect or incomprehensible; results unanalyzed.

Criterion	Pts	Strong	Weak
Results & comparison	3	Baseline comparisons visible; tables/figures readable; main finding highlighted clearly.	No baselines; unreadable visuals; no clear takeaway.
Q&A depth	1	All team members engage questions; answers show understanding of work and limits.	Only one member answers; deflects questions; unaware of limitations.
Slide quality & timing	1	Within 15-minute limit (+/- 1 min); no wall-of-text slides; figures cited.	Over or under time by >2 min; dense text; uncited figures.

10. Peer Review Process & Form

Each team reviews one other team's complete draft (submitted Nov 15). Assignments are posted on Canvas by Nov 16. The peer review is due Nov 22 and is completed independently outside of class using the rubric above. Reviews are graded on specificity, constructiveness, and completeness, not on agreement with the instructor.

CAI 6107 -- Peer Review Form	
Paper ID reviewed	
Reviewing team	
Date submitted	
Summary (3-4 sentences)	Summarize the paper's problem, approach, and main result in your own words. Do NOT copy the abstract.
Strengths (2-3 specific)	What does the paper do well? Be specific, cite section numbers or results.
Weaknesses (2-3 specific)	What are the most significant technical or writing problems? Be constructive.
Actionable suggestions	List 2-3 concrete changes the authors can make before the final submission.
Experiments assessment	Are baselines fair? Are metrics appropriate? Is the protocol reproducible? What is missing?
Notebook assessment	Does the notebook reproduce the paper's results? Are all cells executed? Is the code commented?
Writing & clarity	Which sections need the most revision? Any unclear terminology or missing definitions?
Overall recommendation	Strong accept / Accept / Weak accept / Borderline / Reject, with one-sentence justification.
Confidential note (optional)	Anything for the instructor that you prefer not to share with the authors.

11. AI Tool Policy

Permitted

- GitHub Copilot in VS Code for all implementation and coding tasks.
- AI writing assistants (Claude, ChatGPT, Grammarly) for grammar checking and sentence-level clarity.
- AI literature search tools (Semantic Scholar, Elicit, Connected Papers) for finding related work.

Not permitted

- AI generation of the research question, hypothesis, or core contribution.
- AI generation of the experimental design or methodology.
- AI generation of complete paper sections without substantial human revision and intellectual ownership.

AI Usage Statement (required)

Every submission from M3 onward must include an AI Usage Statement appended after the references. Example:

> *"GitHub Copilot was used for Python implementation. Claude was used to suggest grammar edits in Sections 1 and 3. All research design, experimental setup, analysis, and conclusions are the authors' own work."*

Submissions without an AI Usage Statement will be returned ungraded.

12. Submission Instructions

All milestone submissions are made via Canvas. File naming convention:

- M1: TeamName_M1.pdf
- M2: TeamName_M2.pdf
- M3-M5: TeamName_M3.pdf + TeamName_M3_source.zip (LaTeX source) + Paper should be available on Overleaf
- Peer review: TeamName_Review_of_TeamX.pdf

Final submission (all five items required):

- TeamName_Final.pdf (compiled paper)
- TeamName_Final_source.zip (LaTeX source)
- TeamName_notebook.ipynb (fully executed Jupyter Notebook)
- TeamName_notebook.pdf (PDF export of executed notebook)
- TeamName_data.zip (dataset + README.txt)

Missing any item from the final submission results in a 3-point deduction per missing item from the project grade.

13. Project Calendar

Date	Event
Aug 22	M1 due -- Team registration + topic area (2 pts)
Sep 06	M2 due -- Research proposal, 2 pages (3 pts)
Sep 09	Proposal feedback returned by instructor
Oct 04	M3 due -- Introduction & Related Work in FLAIRS LaTeX (5 pts)
Oct 25	M4 due -- Methodology & Experiments draft in LaTeX (5 pts)
Nov 15	M5 due -- Complete paper draft, all sections, FLAIRS format (5 pts), submitted for peer review
Nov 16	Peer review assignments posted on Canvas
Nov 22	Peer review due (5 pts), completed independently outside class
Nov 30	FINAL SUBMISSION DUE + Project Presentations (single session, Final Exam Slot) -- submit all five items to Canvas before 3:00 pm
Dec 1	Final Presentation
Jan 22, 2027	FLAIRS-40 abstract deadline, voluntary submission for teams continuing with instructor (not a course requirement)
Feb 01, 2027	FLAIRS-40 full paper deadline, voluntary submission for teams continuing with instructor (not a course requirement)

Questions about the project should be raised during class or office hours. Start early. The most common mistake is underestimating how long it takes to run experiments, debug code, and write a coherent paper. A late start on M3 almost always results in a weak final submission.